Laboratory Work 3

Active Directory Deployment

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Deployment platform: Oracle VirtualBox

Objective and report scope

The objective is to deploy the identity-lab.local Active Directory domain for Northstar Labs, configure DNS and directory objects, and prepare the environment for a later Keycloak LDAP integration. This report documents virtual machine preparation, network configuration and troubleshooting, and installation of the AD DS and DNS Server roles. Domain creation and validation will be documented as the work progresses.

1 Initial virtual machine inventory

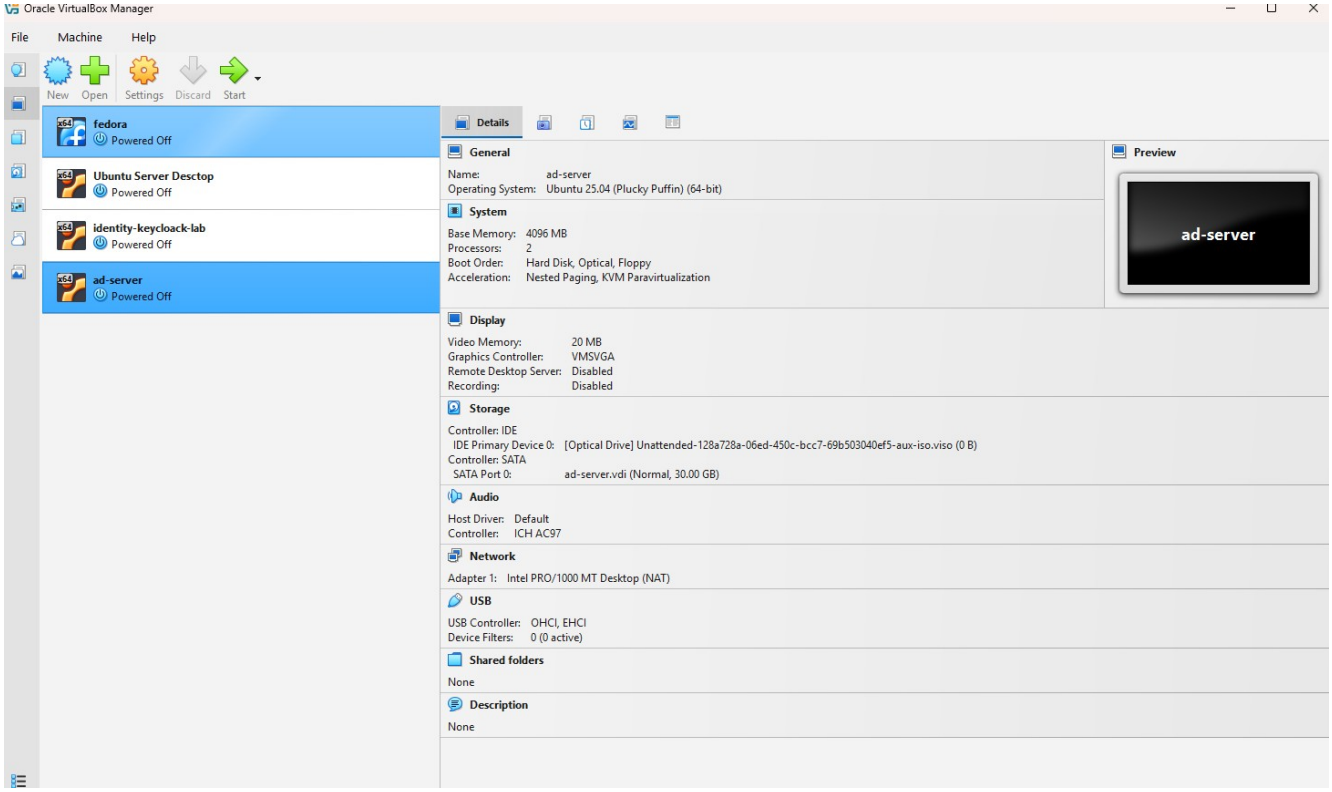


Figure 1. Initial VirtualBox environment

I reviewed the virtual machines available in VirtualBox, including identity-keycloak-lab and ad-server. At this stage, ad-server was powered off and its configured OS profile was Ubuntu 25.04, with 4096 MB RAM, two processors, and a 30 GB virtual disk; this screenshot records the initial inventory and does not confirm a Windows Server installation.

2 Host only network preparation

The laboratory uses a shared host-only network for communication between the virtual machines and the host computer. Each server also uses a NAT adapter for outbound internet access.

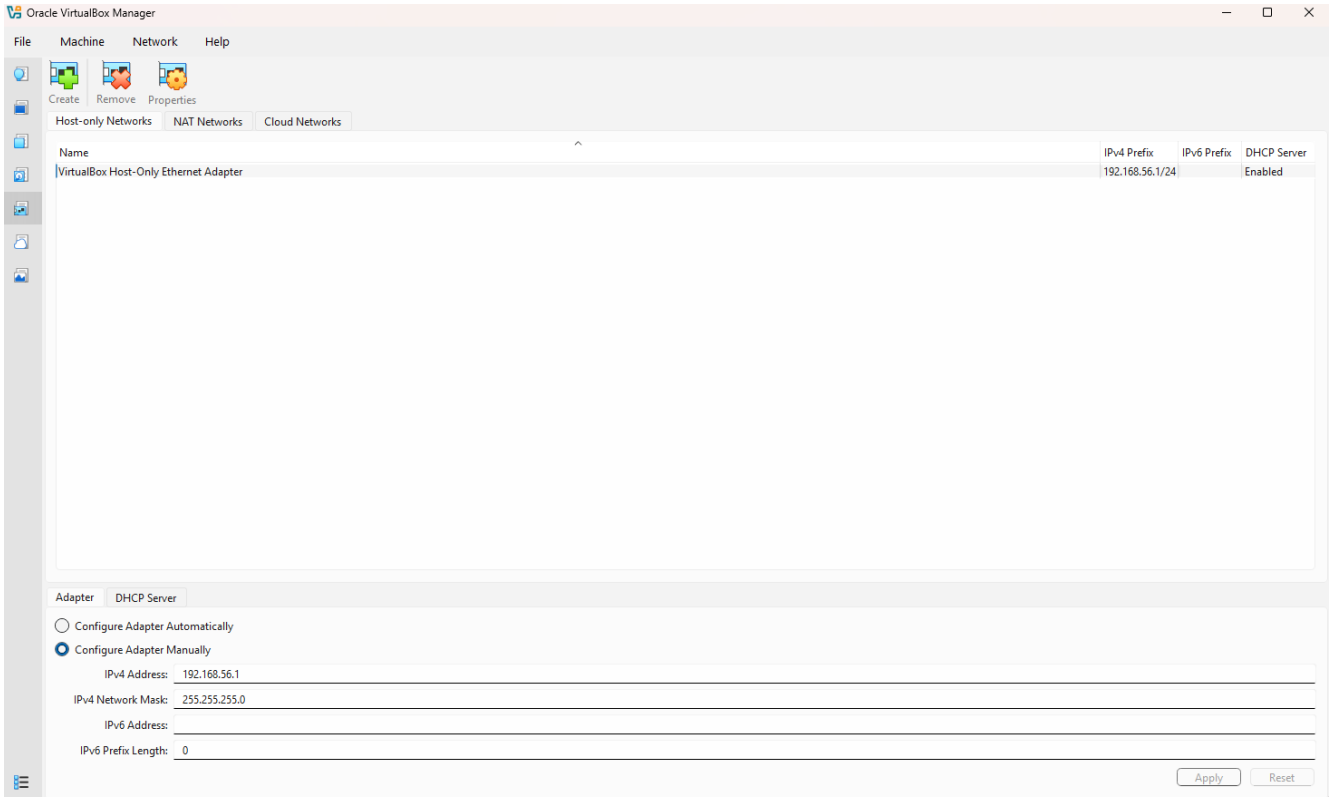


Figure 2. VirtualBox host-only network settings

I verified the VirtualBox Host-Only Ethernet Adapter with host address 192.168.56.1 and subnet mask 255.255.255.0. The network was present and DHCP was enabled, providing the shared 192.168.56.0/24 network required for the local laboratory.

3 Keycloak virtual machine adapters

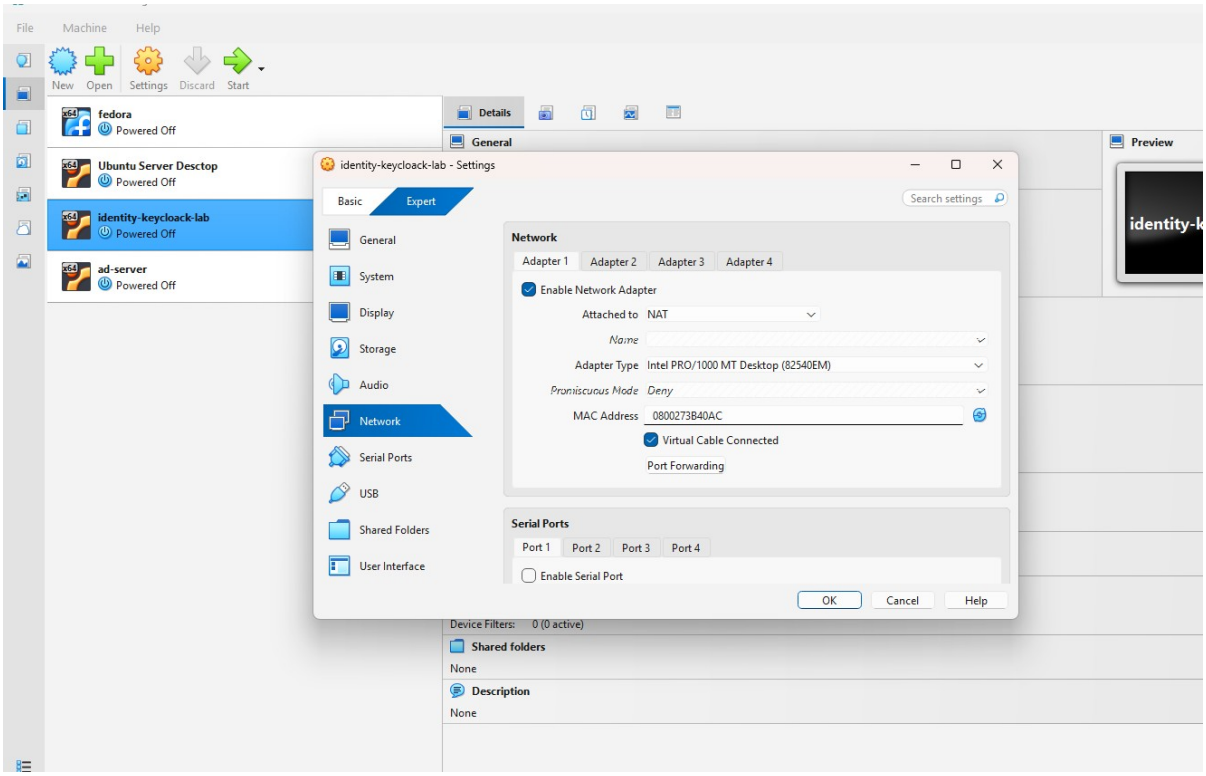


Figure 3. Keycloak Adapter 1 configured for NAT

I configured Adapter 1 of identity-keycloak-lab to use NAT and kept the virtual cable connected. The displayed settings confirm that the adapter was enabled for outbound internet connectivity.

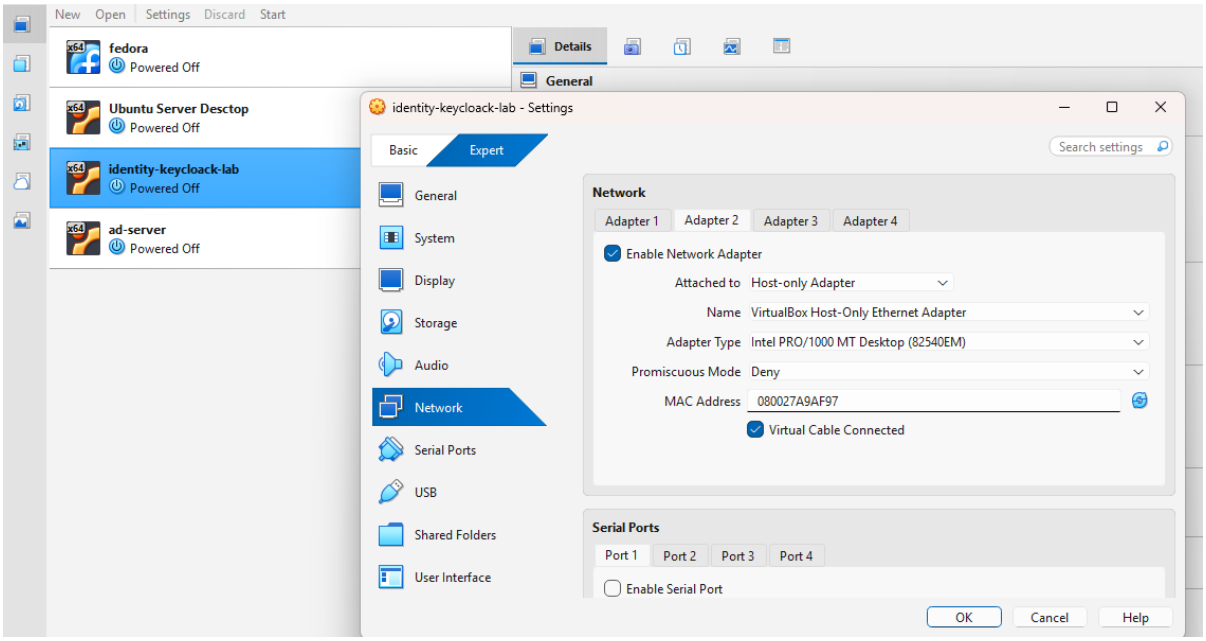


Figure 4. Keycloak Adapter 2 configured for the host-only network

I enabled Adapter 2 and attached it to the VirtualBox Host-Only Ethernet Adapter with the virtual cable connected. This connects the virtual hardware to the laboratory network; the guest operating system interface still requires configuration.

4 Keycloak guest network inspection

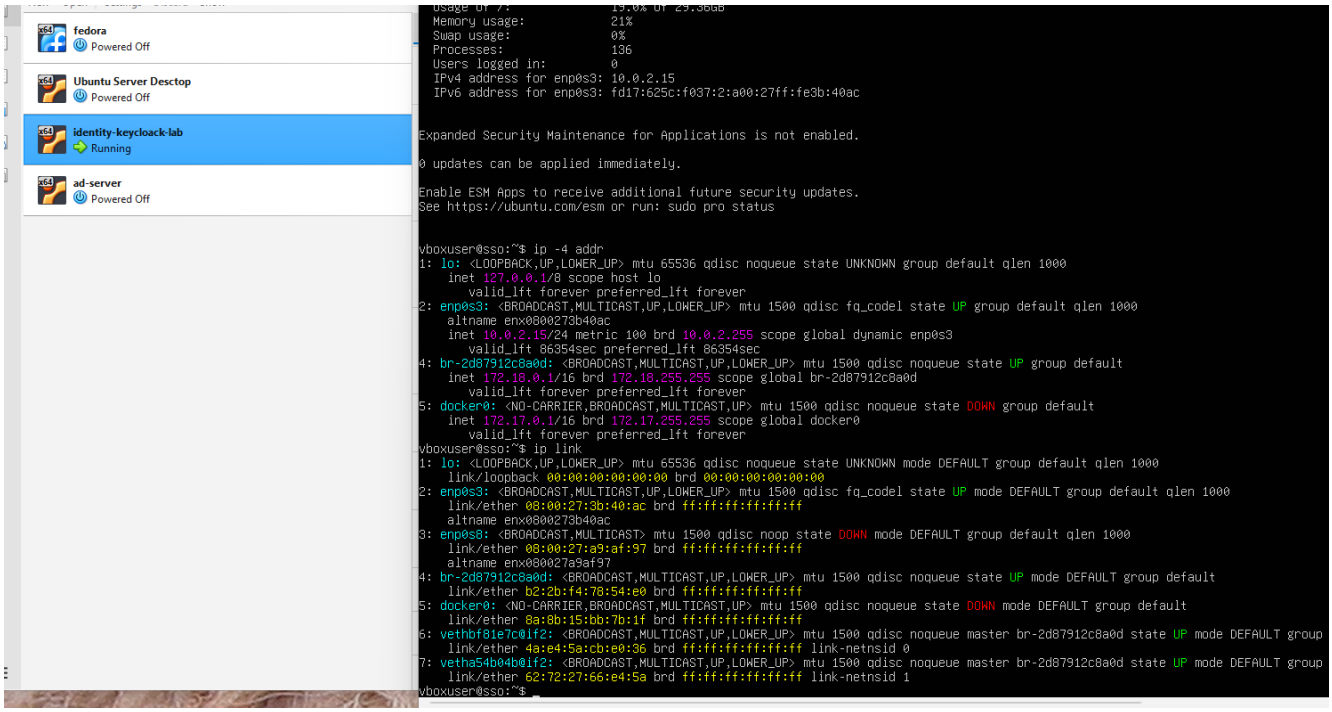


Figure 5. Ubuntu interface and IPv4 address inspection

I ran ip -4 addr and ip link to inspect the Keycloak VM network interfaces. The NAT interface enp0s3 was UP with address 10.0.2.15/24, while enp0s8 was DOWN and had no displayed IPv4 address, so host-only connectivity was not yet established.

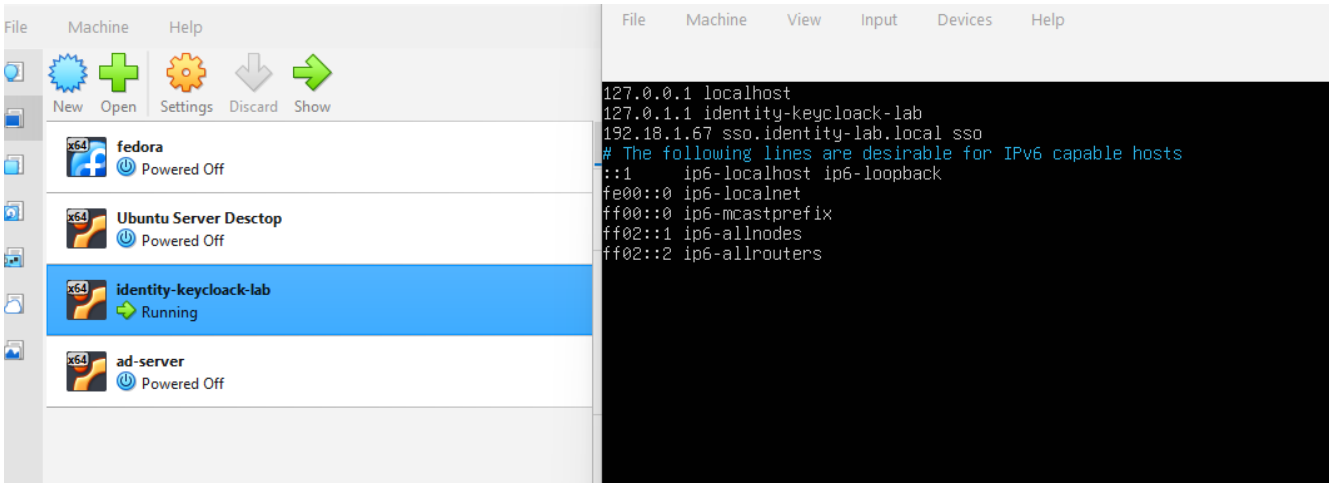


Figure 6. Existing local hostname mapping

I inspected /etc/hosts and found the entry 192.18.1.67 sso.identity-lab.local sso. This address differs from the planned host-only subnet, so the mapping needs to be updated after the Keycloak host-only address is configured.

5 Windows Server virtual machine network settings

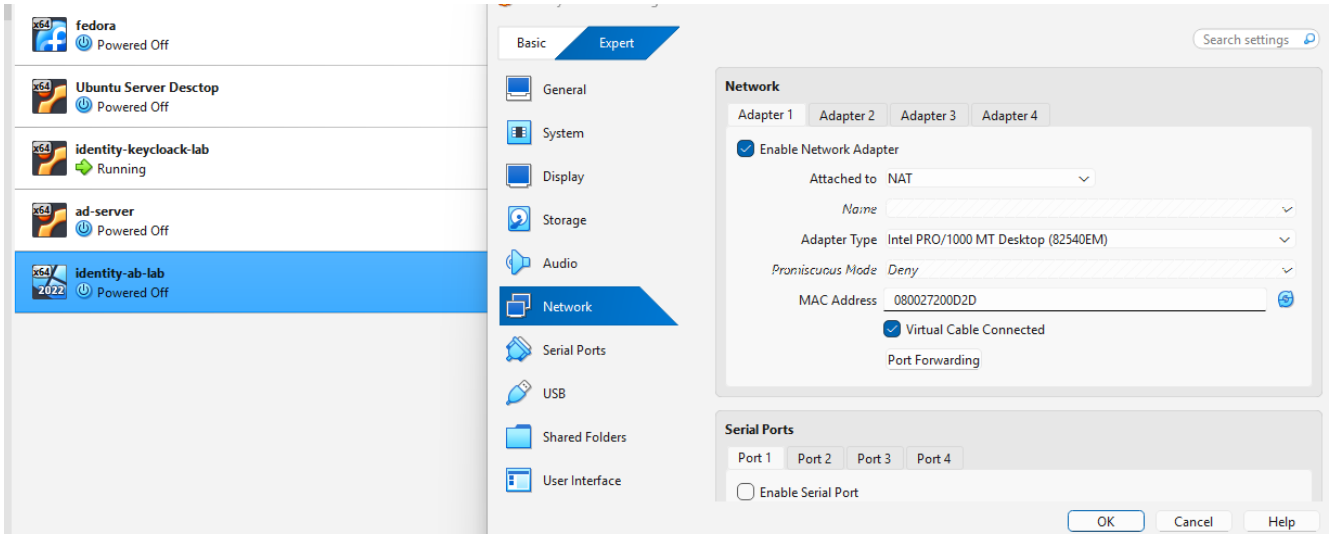


Figure 7. NAT adapter for identity-ab-lab

I configured Adapter 1 of the newly listed identity-ab-lab virtual machine to use NAT with the virtual cable connected. The VM was powered off, and this screenshot confirms its adapter settings rather than completion of the Windows Server installation.

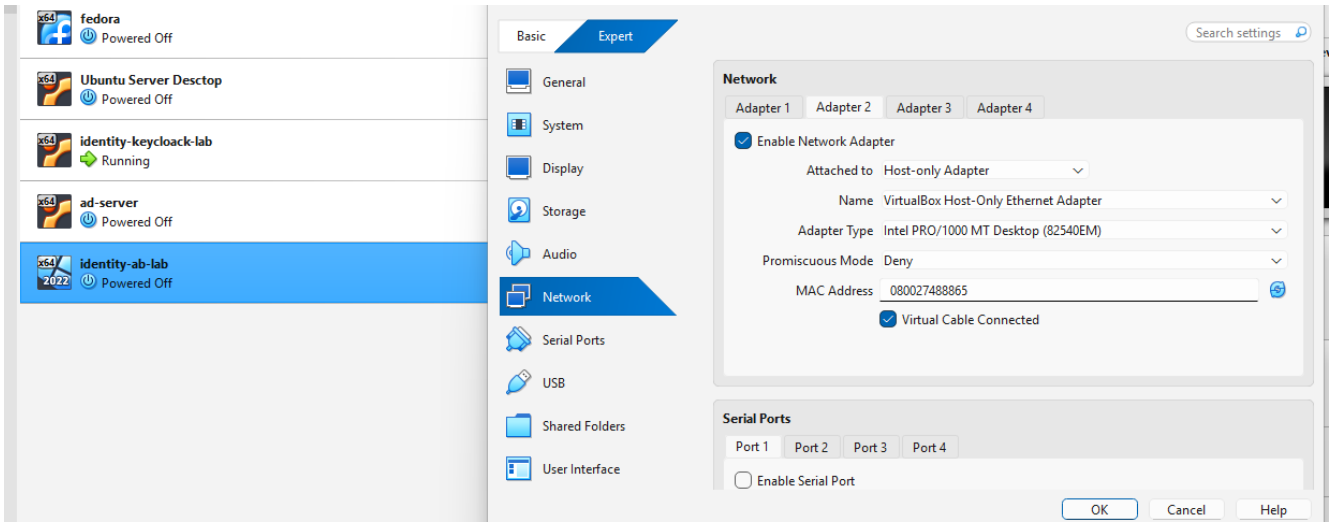


Figure 8. Host-only adapter for identity-ab-lab

I enabled Adapter 2 and selected the same VirtualBox Host-Only Ethernet Adapter used by the Keycloak VM. Both virtual machines now have matching host-only adapter selections, while guest IP configuration and communication tests remain to be documented.

6 Windows network and connectivity checks

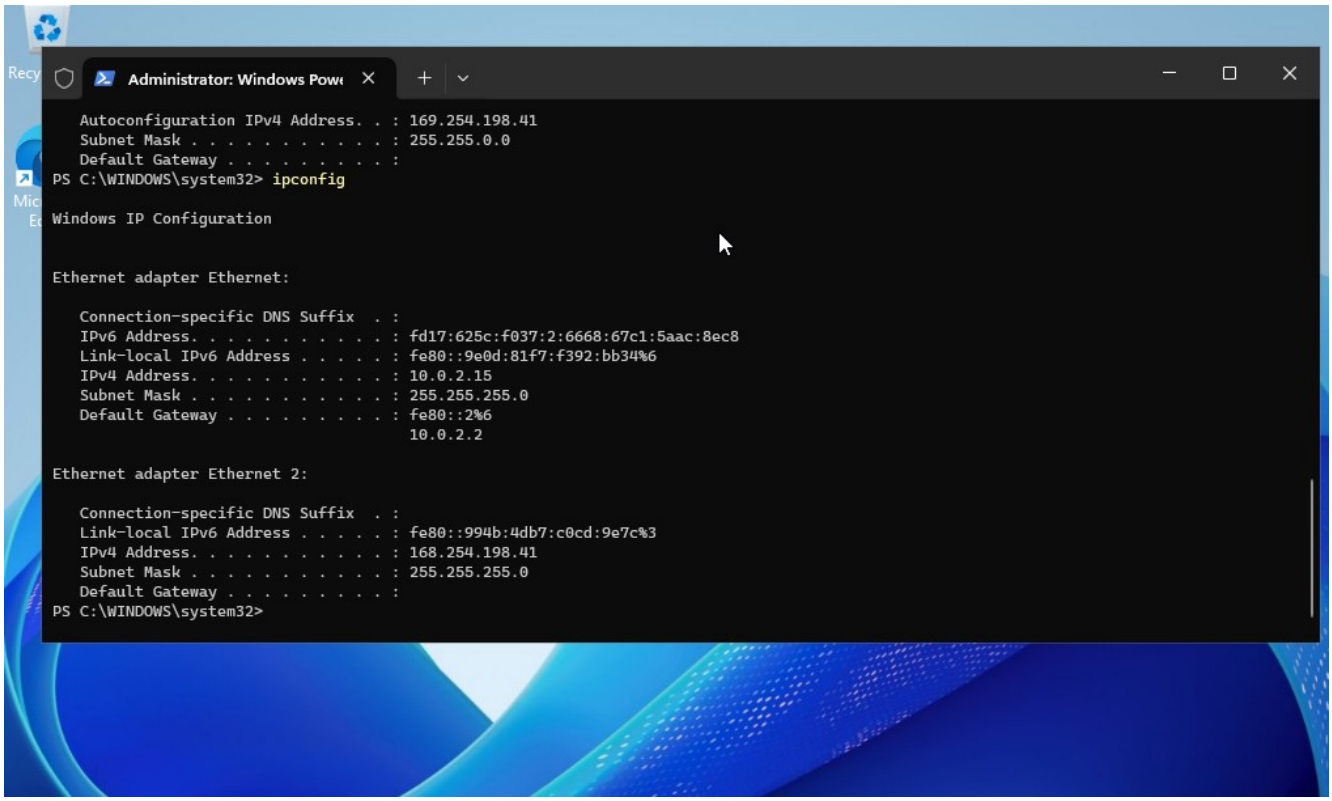


Figure 9. Windows IPv4 configuration inspection

I ran ipconfig in an elevated PowerShell window to inspect the Windows network configuration. Ethernet used 10.0.2.15/24 with gateway 10.0.2.2, while Ethernet 2 displayed 168.254.198.41/24 without a default gateway. The second adapter address was outside the intended 192.168.56.0/24 host-only subnet, so its configuration still needed correction.

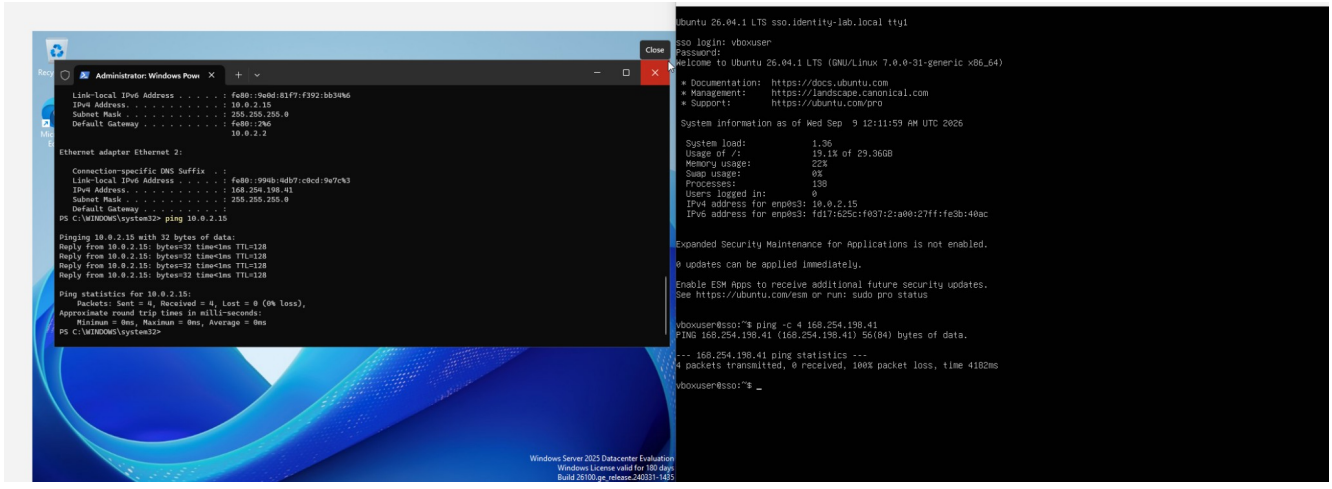


Figure 10. Initial ping tests on Windows Server and Ubuntu

I ran ping 10.0.2.15 on Windows Server 2025 Datacenter Evaluation and ping -c 4 168.254.198.41 on the Ubuntu Keycloak VM. Windows received four replies from its own NAT address, while Ubuntu received no replies and reported 100% packet loss. These results did not establish connectivity between the two VMs, so further host-only network configuration was required.

7 Host only network verification during troubleshooting

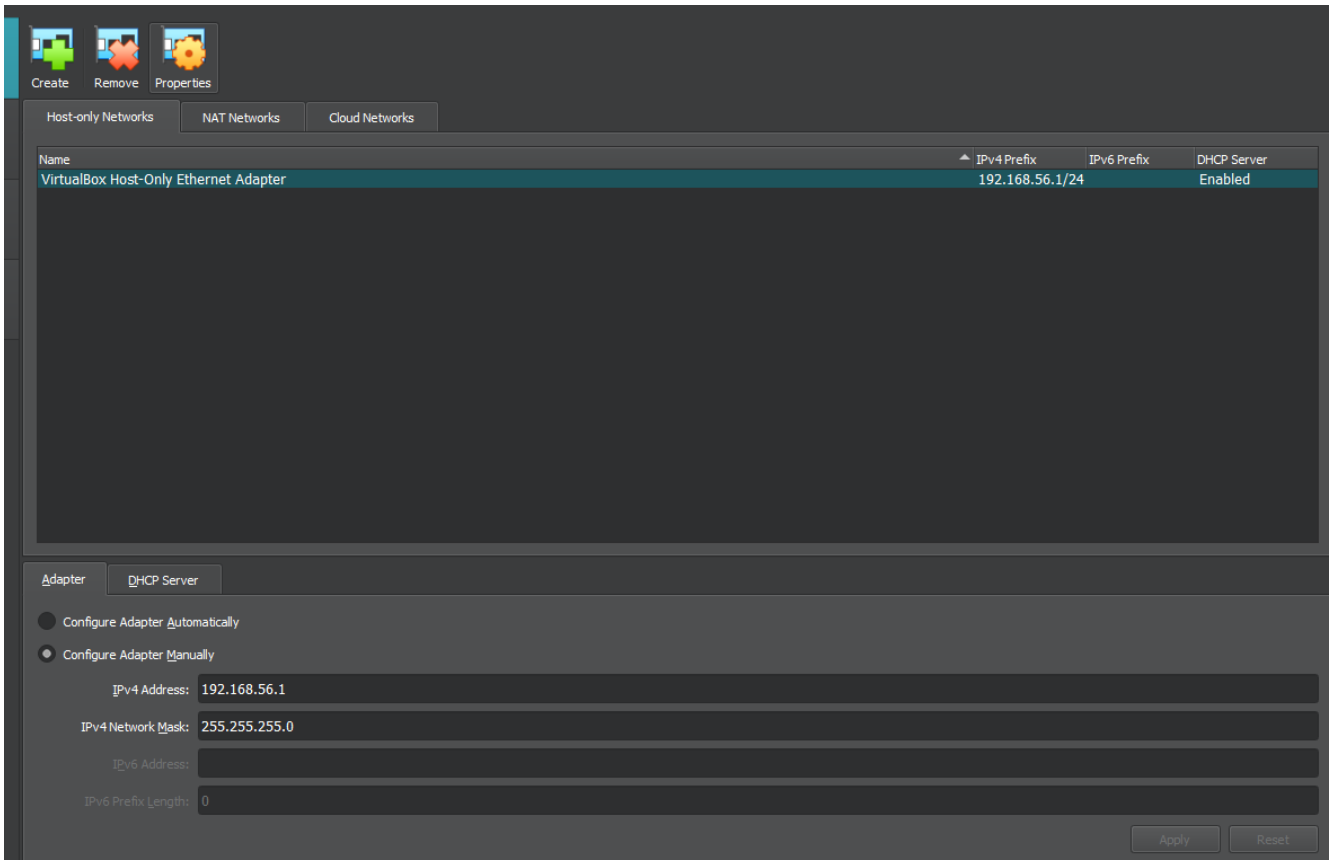


Figure 11. Rechecking the VirtualBox host-only subnet

I reopened VirtualBox Network Manager and verified that the host-only adapter used 192.168.56.1 with subnet mask 255.255.255.0 and DHCP enabled. This confirmed the intended 192.168.56.0/24 subnet and highlighted the mismatch with the Windows Ethernet 2 address shown in Figure 9.

8 Keycloak interface and routing inspection

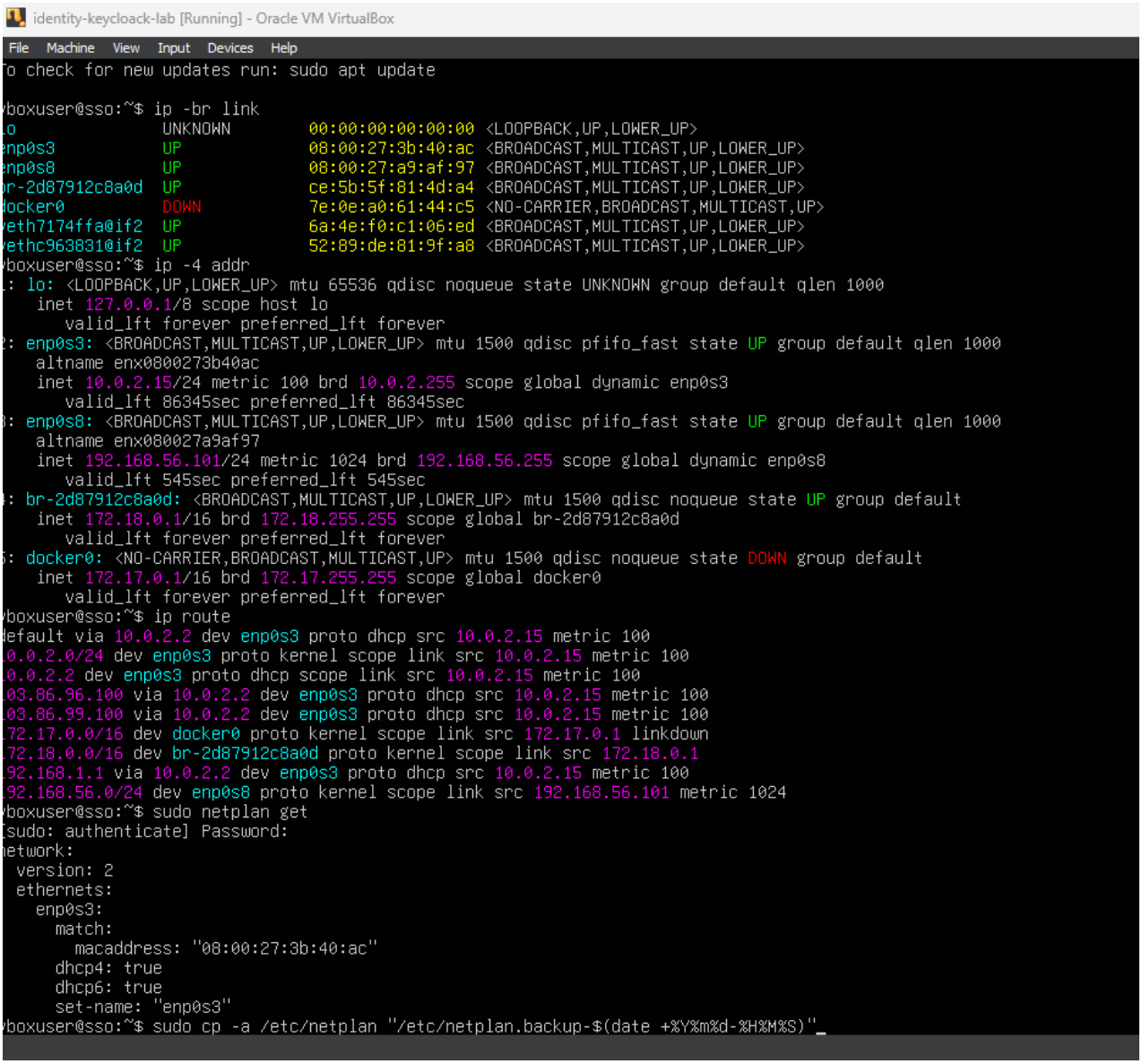


Figure 12. Inspecting active interfaces and the existing Netplan configuration

I inspected the interfaces, IPv4 addresses, routing table, and Netplan configuration using ip -br link, ip -4 addr, ip route, and sudo netplan get. The host-only interface enp0s8 was now UP with the dynamic address 192.168.56.101/24, while the default route remained through the NAT interface enp0s3. The displayed Netplan configuration listed only enp0s3, so a persistent static configuration for enp0s8 was still needed.

9 Static address configuration for Keycloak

```
vboxuser@sso:~$ sudo chmod 600 /etc/netplan/99-identity-lab.yaml
vboxuser@sso:~$ sudo netplan get
network:
  version: 2
  ethernets:
    enp0s3:
      match:
        macaddress: "08:00:27:3b:40:ac"
      dhcp4: true
      dhcp6: true
      set-name: "enp0s3"
    enp0s8:
      optional: true
      addresses:
      - "192.168.56.10/24"
      dhcp4: false
vboxuser@sso:~$ sudo netplan generate
vboxuser@sso:~$ sudo netplan try
Do you want to keep these settings?


Press ENTER before the timeout to accept the new configuration


Changes will revert in 116 seconds
Configuration accepted.
vboxuser@sso:~$ _
```

Figure 13. Applying and accepting the Netplan configuration

I set the permissions of /etc/netplan/99-identity-lab.yaml to 600 and verified that Netplan configured enp0s8 with 192.168.56.10/24, dhcp4 disabled, and optional enabled, while retaining DHCP on enp0s3. I ran sudo netplan generate and sudo netplan try, then accepted the new settings. The terminal reported Configuration accepted, confirming acceptance of the network configuration.

```
 Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 47
vboxuser@sso:~$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    altname enx0800273b40ac
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 86250sec preferred_lft 86250sec
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    altname enx080027a9af97
    inet 192.168.56.10/24 brd 192.168.56.255 scope global enp0s8
       valid_lft forever preferred_lft forever
4: br-2d87912c8a0d: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-2d87912c8a0d
       valid_lft forever preferred_lft forever
5: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
vboxuser@sso:~$ _
```

Figure 14. Verifying the static host-only IPv4 address

I ran ip -4 addr after applying the Netplan configuration. The output confirmed that enp0s8 was UP with 192.168.56.10/24 and unlimited address lifetimes, while enp0s3 retained its dynamic NAT address of 10.0.2.15/24. The Keycloak host-only address was therefore configured as planned.

10 Windows Server name and adapter verification

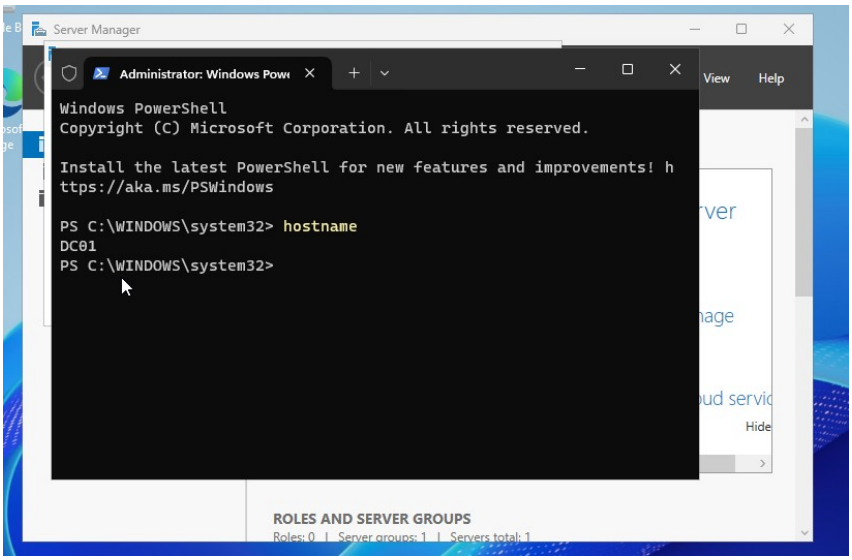


Figure 15. Verifying the Windows Server hostname

I ran hostname in an elevated PowerShell window to verify the server name. The command returned DC01, confirming that the Windows Server computer name matched the laboratory requirement.

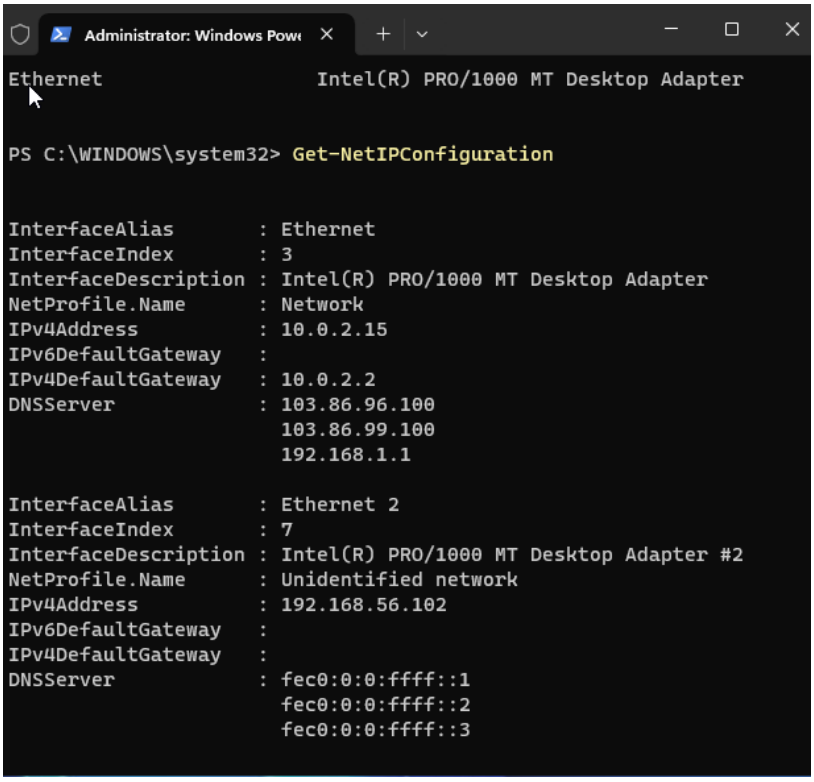


Figure 16. Inspecting Windows network adapters before static addressing

I ran Get-NetIPConfiguration to inspect both Windows network interfaces. Ethernet used the NAT address 10.0.2.15 with gateway 10.0.2.2, while Ethernet 2 used 192.168.56.102 without an IPv4 default gateway. The host-only adapter was now in the intended subnet, but its planned static address and DNS settings still needed to be applied.

11 Static server addressing and successful connectivity

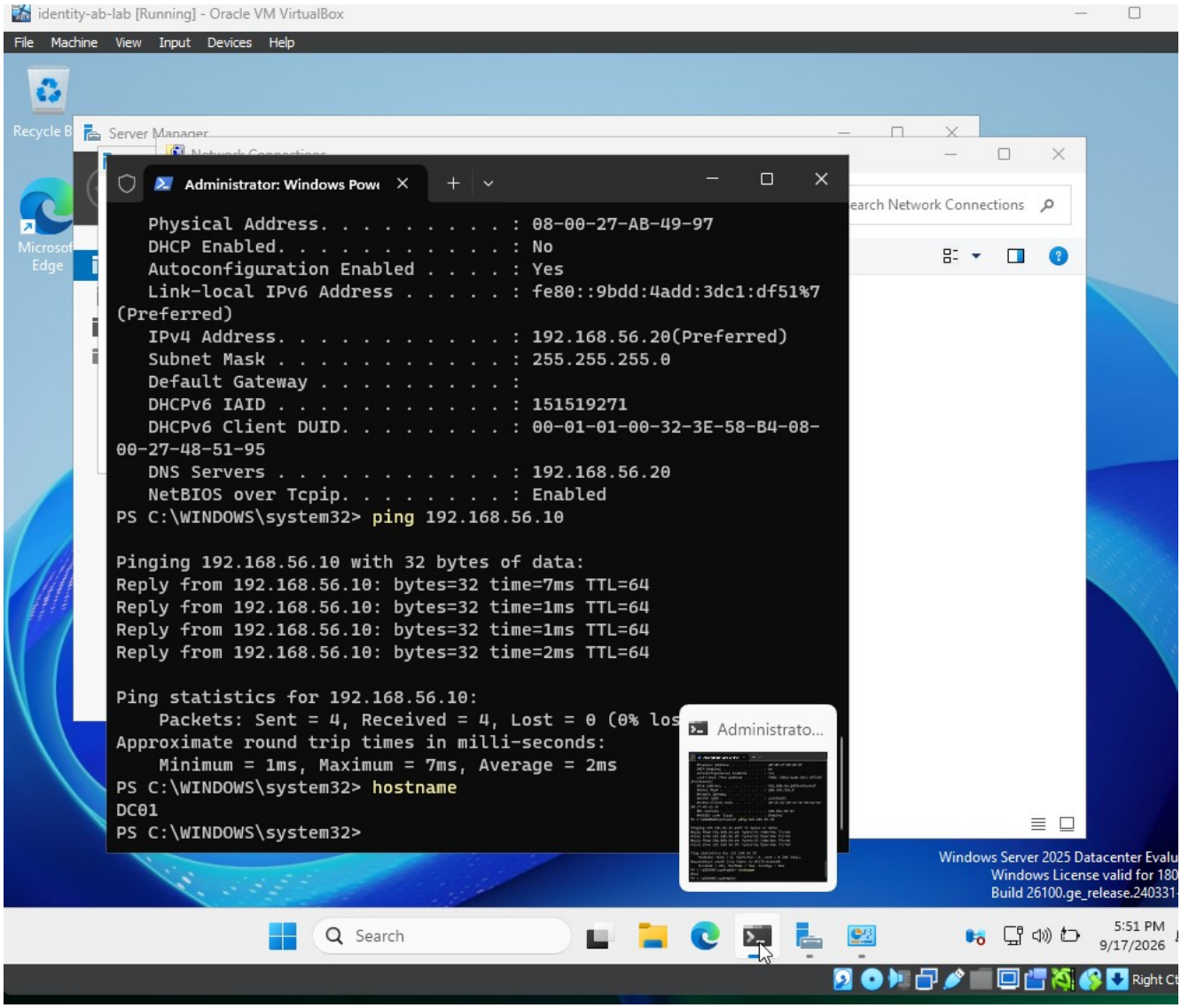


Figure 17. Confirming the static server address and ping to Keycloak

I configured the Windows host-only adapter with static IPv4 address 192.168.56.20, subnet mask 255.255.255.0, and DNS server 192.168.56.20, leaving the default gateway blank; the displayed configuration confirmed that DHCP was disabled. I then ran ping 192.168.56.10 and received four replies with 0% packet loss, confirming connectivity from Windows Server to the Keycloak VM. A subsequent hostname command returned DC01.

12 Installation of AD DS and DNS Server roles

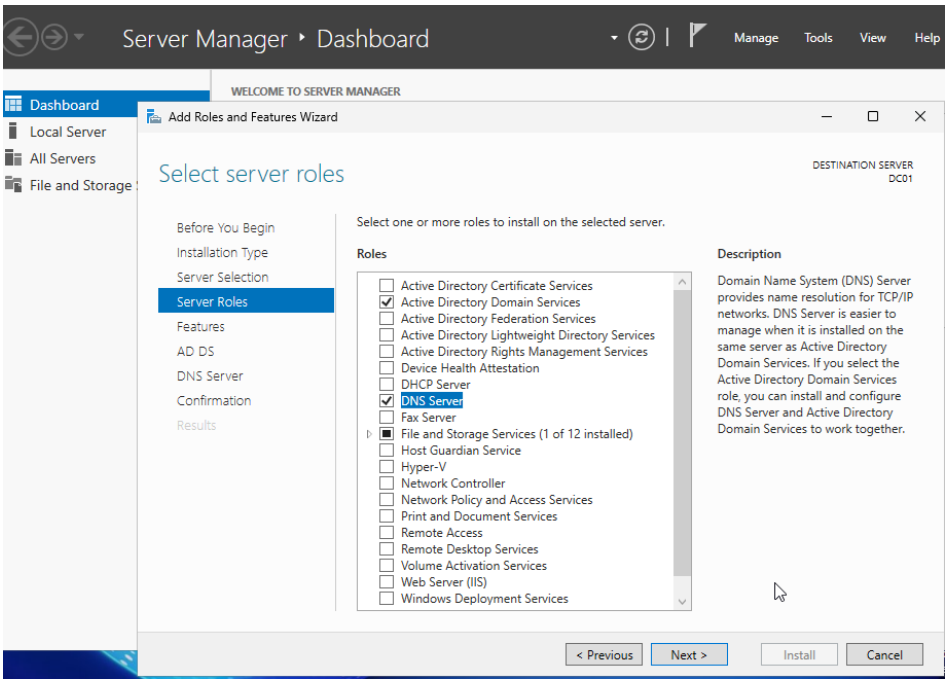


Figure 18. Selecting AD DS and DNS Server on DC01

I opened the Add Roles and Features Wizard in Server Manager and selected Active Directory Domain Services and DNS Server for DC01. Both roles were checked on the Server Roles page, preparing the required components for installation.

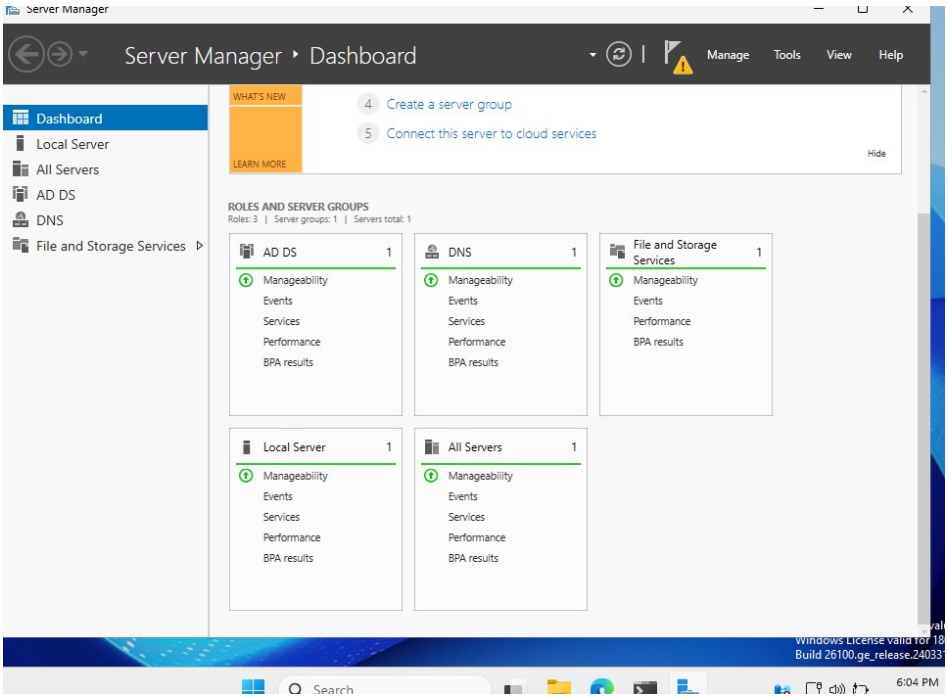
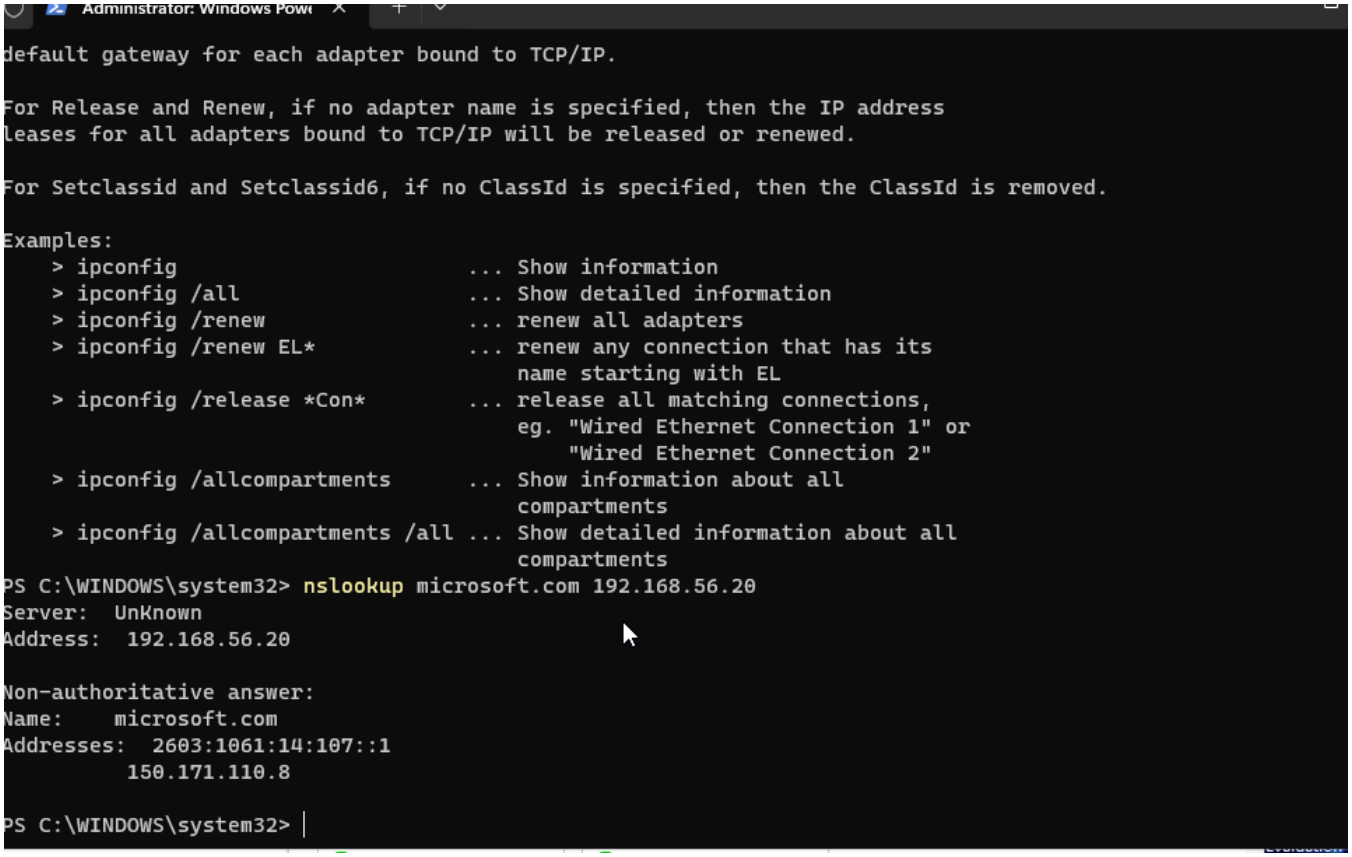
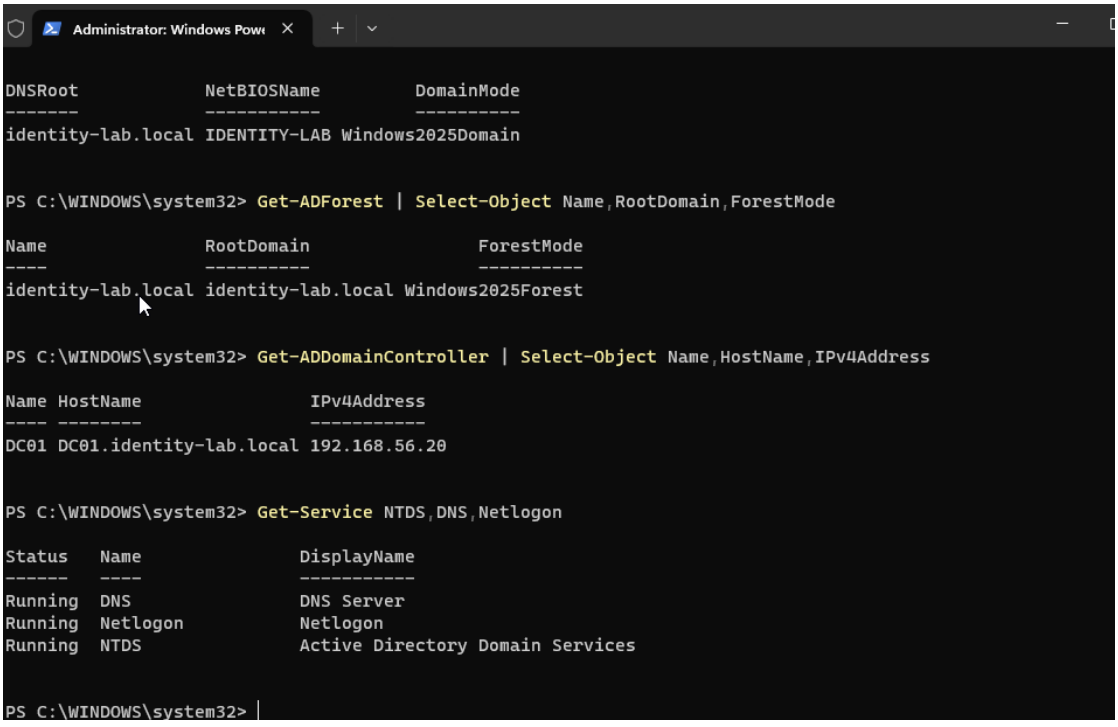


Figure 19. Verifying the installed roles in Server Manager

After installing the selected roles, I verified that AD DS and DNS appeared in the Server Manager navigation pane and dashboard. Both role tiles displayed green manageability indicators, and a notification flag remained visible.

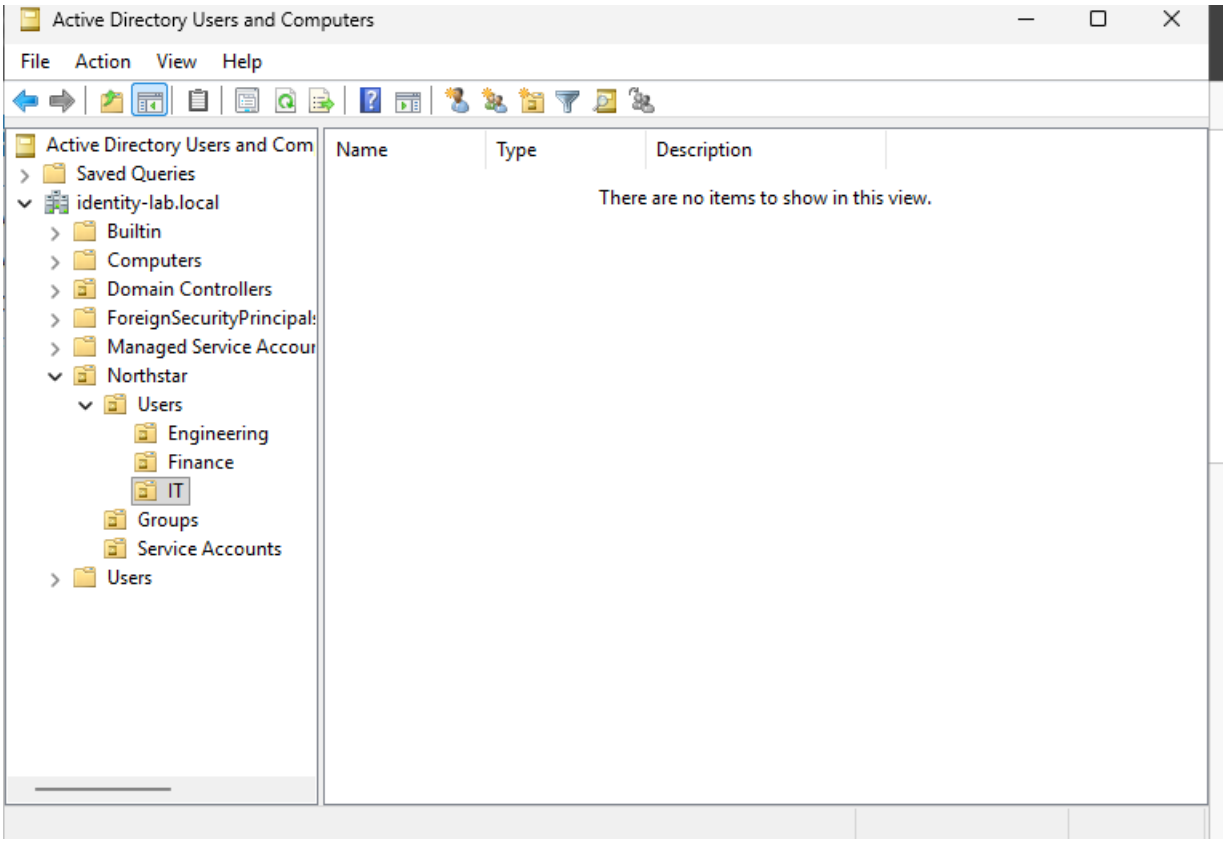
The roles were available for configuration, while promotion to a domain controller and creation of identity-lab.local still needed to be documented.

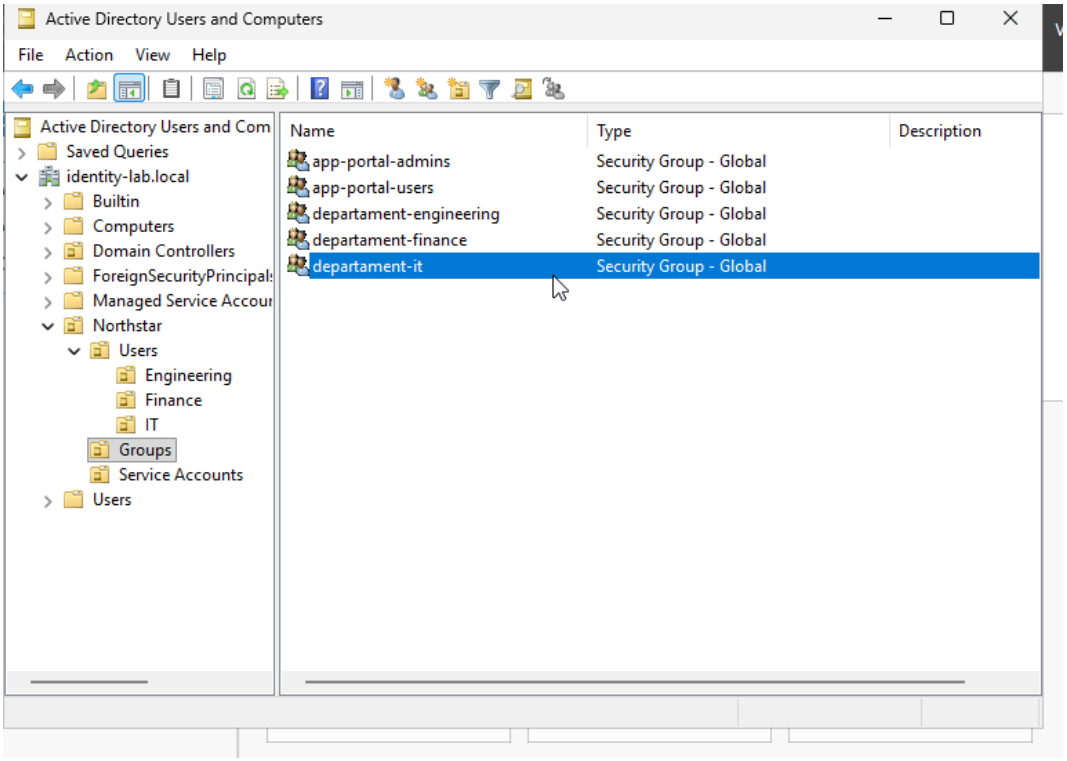


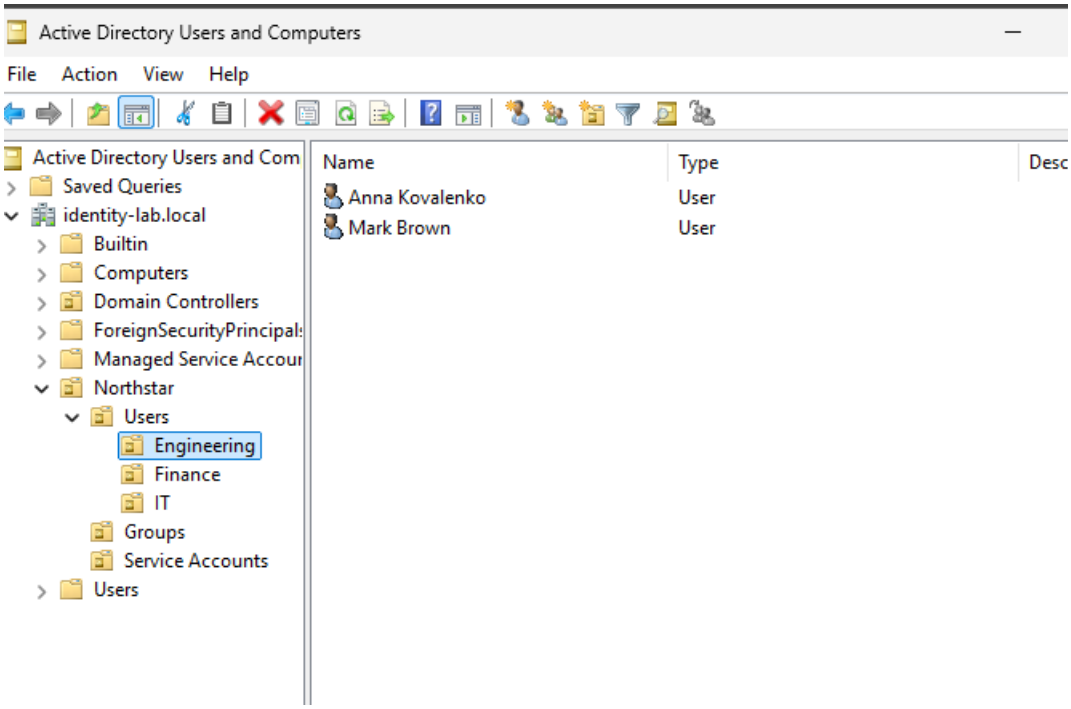


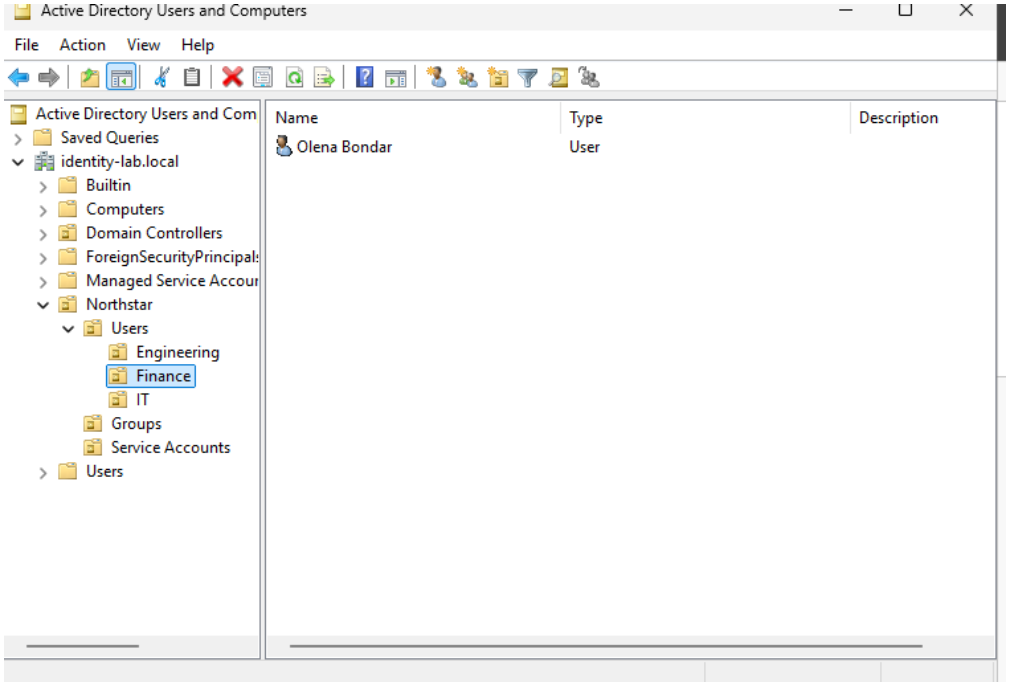
```
PS C:\WINDOWS\system32> nslookup -type=SRV _ldap._tcp.dc._msdcs.identity-lab.local 192.168.56.20
Server:  UnKnown
Address:  192.168.56.20

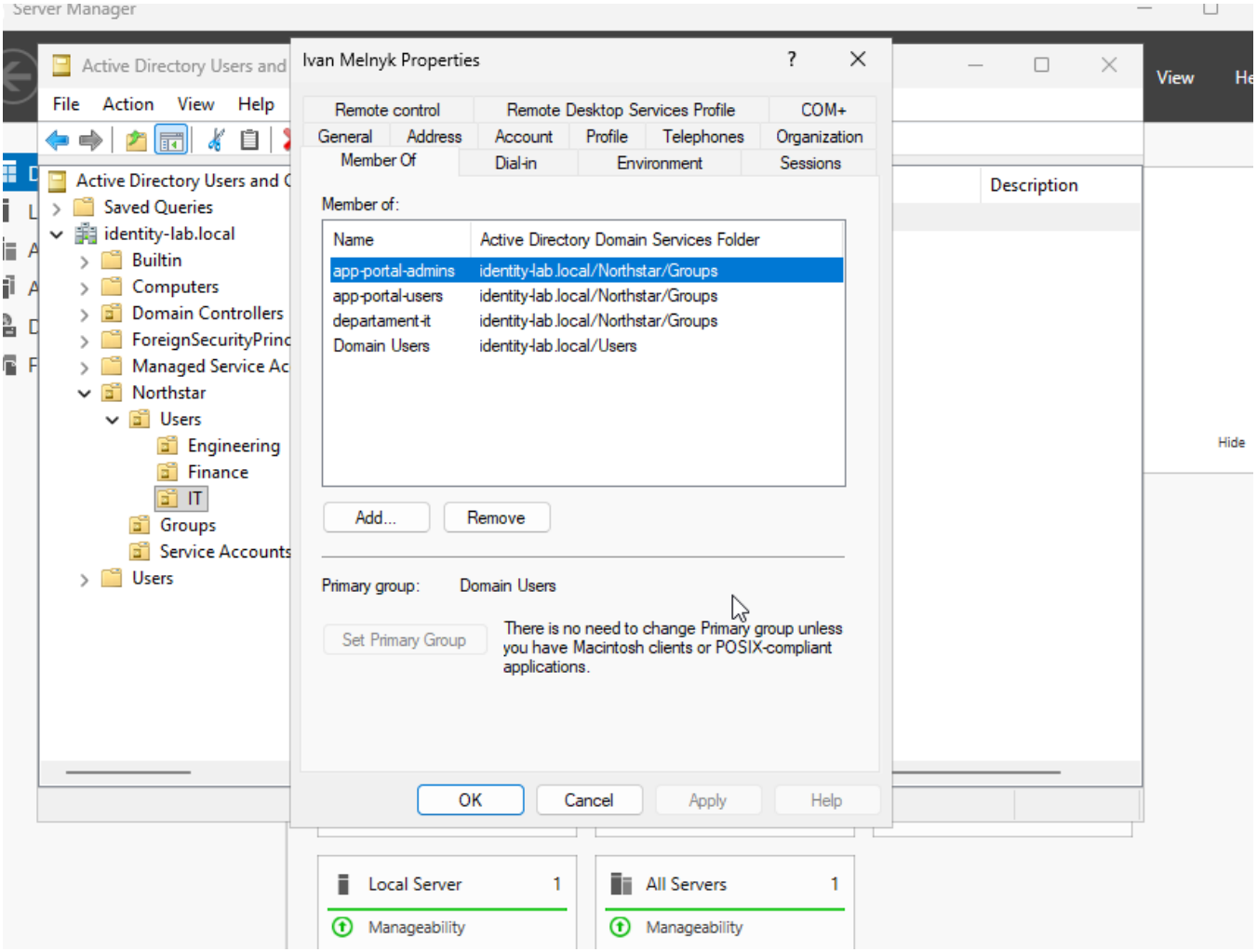
_ldap._tcp.dc._msdcs.identity-lab.local SRV service location:
          priority       = 0
          weight         = 100
          port           = 389
          svr hostname   = dc01.identity-lab.local
dc01.identity-lab.local internet address = 192.168.56.20
PS C:\WINDOWS\system32>
```

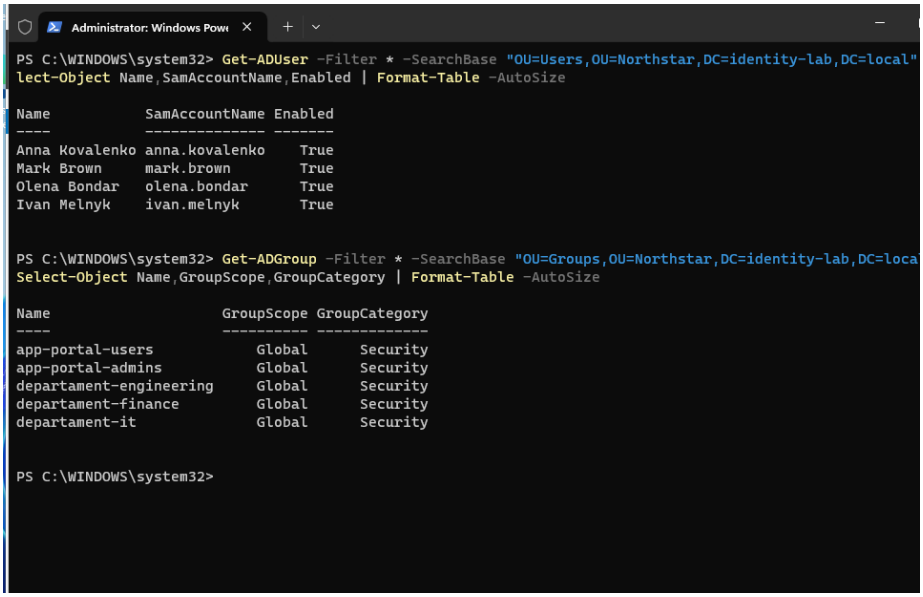



Active Directory Users and Computers
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Saved Queries
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ForeignSecurityPrincipal:
Managed Service Accour
Northstar
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app-portal-users
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departament-finance
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Active Directory Users and Computers
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Anna Kovalenko
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Mark Brown
User


Active Directory Users and Computers
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Server Manager
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Ivan Melnyk Properties
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Organization
Member Of
Dial-in
Environment
Sessions
Member of:
Name
Active Directory Domain Services Folder
app-portal-admins
identity-lab.local/Northstar/Groups
app-portal-users
identity-lab.local/Northstar/Groups
departament-it
identity-lab.local/Northstar/Groups
Domain Users
identity-lab.local/Users
Add...
Remove
Primary group:
Domain Users
Set Primary Group
There is no need to change Primary group unless you have Macintosh clients or POSIX-compliant applications.
OK
Cancel
Apply
Help
Description
Hide
View
Local Server
1
Manageability
All Servers
1
Manageability


Administrator: Windows Powe
PS C:\WINDOWS\system32> Get-ADUser -Filter * -SearchBase "OU=Users,OU=Northstar,DC=identity-lab,DC=local"
lect-Object Name,SamAccountName,Enabled | Format-Table -AutoSize
Name SamAccountName Enabled
Anna Kovalenko anna.kovalenko True
Mark Brown mark.brown True
Olena Bondar olena.bondar True
Ivan Melnyk ivan.melnyk True
PS C:\WINDOWS\system32> Get-ADGroup -Filter * -SearchBase "OU=Groups,OU=Northstar,DC=identity-lab,DC=loca
Select-Object Name,GroupScope,GroupCategory | Format-Table -AutoSize
Name GroupScope GroupCategory
app-portal-users Global Security
app-portal-admins Global Security
departament-engineering Global Security
departament-finance Global Security
departament-it Global Security
PS C:\WINDOWS\system32>

```
lect-Object Name,SamAccountName,Enabled | Format-Table -AutoSize

Name           SamAccountName Enabled
----           -------------- -------
Anna Kovalenko anna.kovalenko    True
Mark Brown     mark.brown        True
Olena Bondar   olena.bondar      True
Ivan Melnyk    ivan.melnyk       True


PS C:\WINDOWS\system32> Get-ADGroup -Filter * -SearchBase "OU=Groups,OU=Northstar,DC=identity-
Select-Object Name,GroupScope,GroupCategory | Format-Table -AutoSize

Name                    GroupScope GroupCategory
----                    ---------- -------------
app-portal-users            Global      Security
app-portal-admins           Global      Security
departament-engineering     Global      Security
departament-finance         Global      Security
departament-it              Global      Security


PS C:\WINDOWS\system32> nslookup sso.identity-lab.local 192.168.56.20
Server:  UnKnown
Address:  192.168.56.20

Name:    sso.identity-lab.local
Address:  192.168.56.10

PS C:\WINDOWS\system32>
```

Manageability | Manageability

```
vboxuser@sso:~$ nslookup dc01.identity-lab.local 192.168.56.20
Server:         192.168.56.20
Address:        192.168.56.20#53

Name:   dc01.identity-lab.local
Address: 192.168.56.20

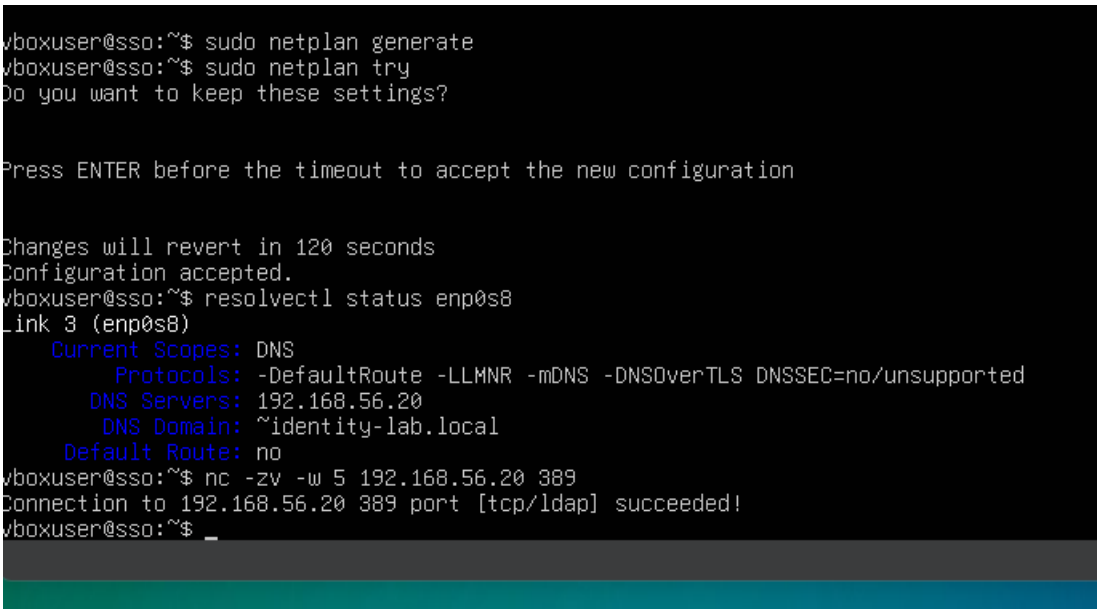
vboxuser@sso:~$ nslookup sso.identity-lab.local 192.168.56.20
Server:         192.168.56.20
Address:        192.168.56.20#53

Name:   sso.identity-lab.local
Address: 192.168.56.10

vboxuser@sso:~$ nslookup -type=SRV _ldap._tcp.dc._msdcs.identity-lab.local 192.168.56.20
Server:         192.168.56.20
Address:        192.168.56.20#53

_ldap._tcp.dc._msdcs.identity-lab.local service = 0 100 389 dc01.identity-lab.local.

vboxuser@sso:~$
```


vboxuser@sso:~$ sudo netplan generate
vboxuser@sso:~$ sudo netplan try
Do you want to keep these settings?
Press ENTER before the timeout to accept the new configuration
Changes will revert in 120 seconds
Configuration accepted.
vboxuser@sso:~$ resolvectl status enp0s8
Link 3 (enp0s8)
Current Scopes: DNS
Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
DNS Servers: 192.168.56.20
DNS Domain: ~identity-lab.local
Default Route: no
vboxuser@sso:~$ nc -zv -w 5 192.168.56.20 389
Connection to 192.168.56.20 389 port [tcp/ldap] succeeded!
vboxuser@sso:~$